

# AI-Driven Customer Retention Optimization for a SaaS Product

## Business Analysis Case Study

### 1. Problem Statement

The SaaS platform is facing high customer churn, particularly during the early stages of the customer lifecycle. This analysis identifies major churn drivers, maps customer behavior patterns, and proposes AI-enabled retention workflows to improve engagement and long-term retention.

### 2. Dataset Used

**Dataset:** Telco Customer Churn Dataset from Kaggle

**Source:** Public customer churn dataset adapted for SaaS behavioral and retention analysis.

**Key Fields Used:** Tenure, Contract Type, Churn Label, Churn Category, Churn Reason, Monthly Charges.

### 3. Project Objectives

- Analyze customer churn behavior using cohort and retention analysis
- Identify high-risk churn segments and lifecycle drop-off patterns
- Conduct root cause analysis for customer attrition
- Design AI-enabled retention workflows and engagement strategies

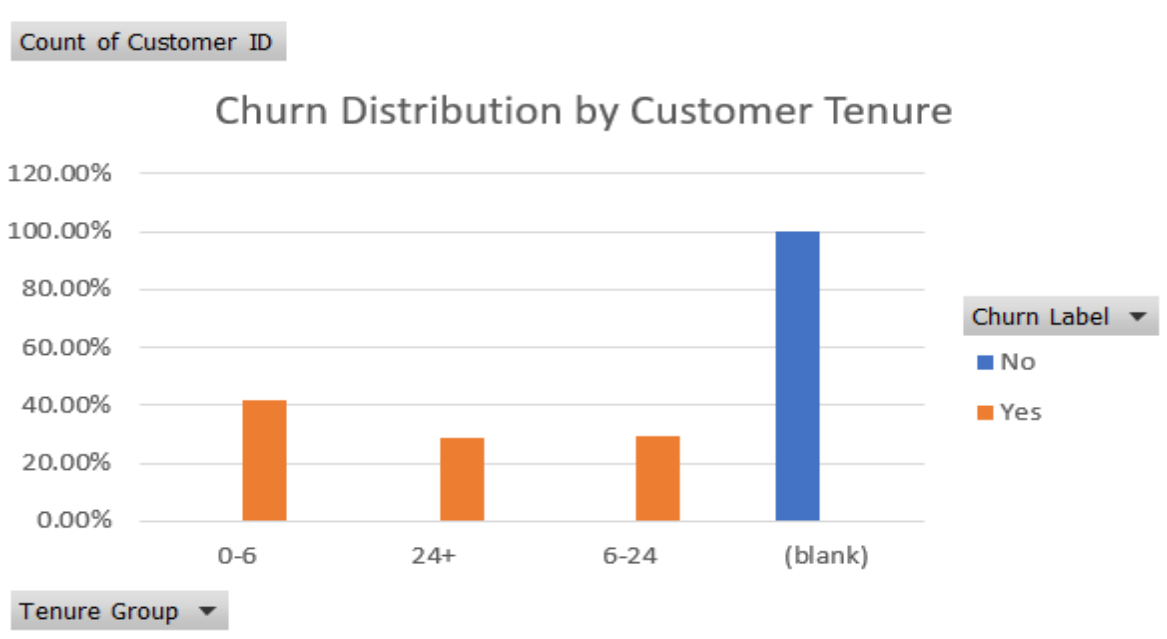
### 4. Methodology

- Cohort segmentation and churn categorization using Excel Pivot Tables
- Customer lifecycle mapping and churn pattern analysis
- Root cause analysis based on churn categories and contract behavior
- AI workflow design for proactive retention intervention

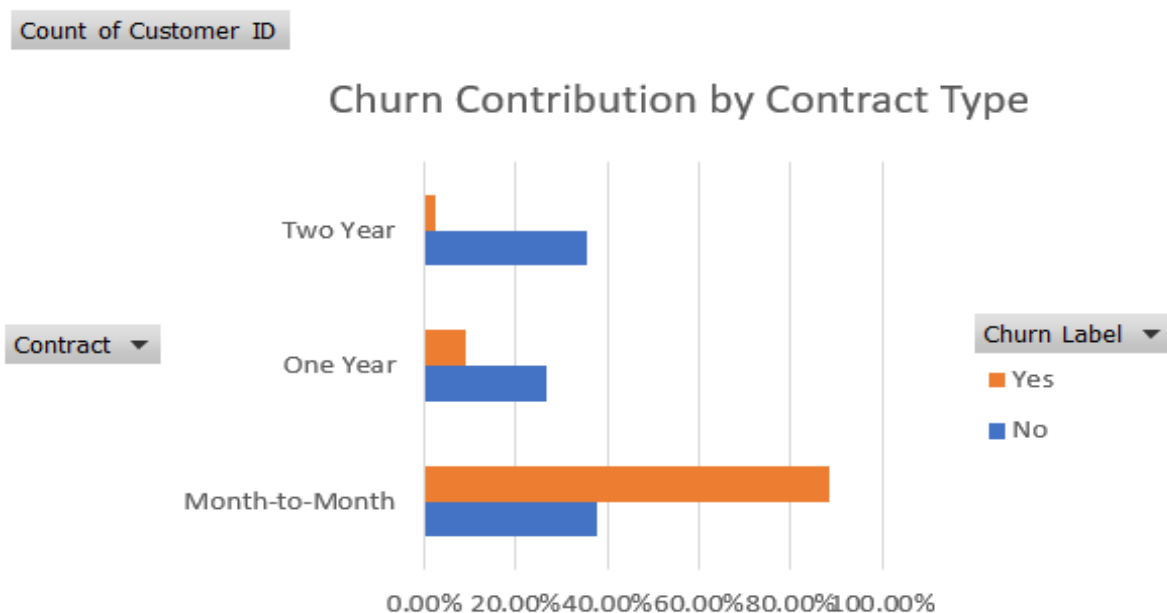
### 5. Key Churn Insights

- 41.95% of total churn occurs within the first 6 months of customer tenure
- 88.55% of churn originates from month-to-month customers
- Competitor-related reasons contribute to 45% of total churn

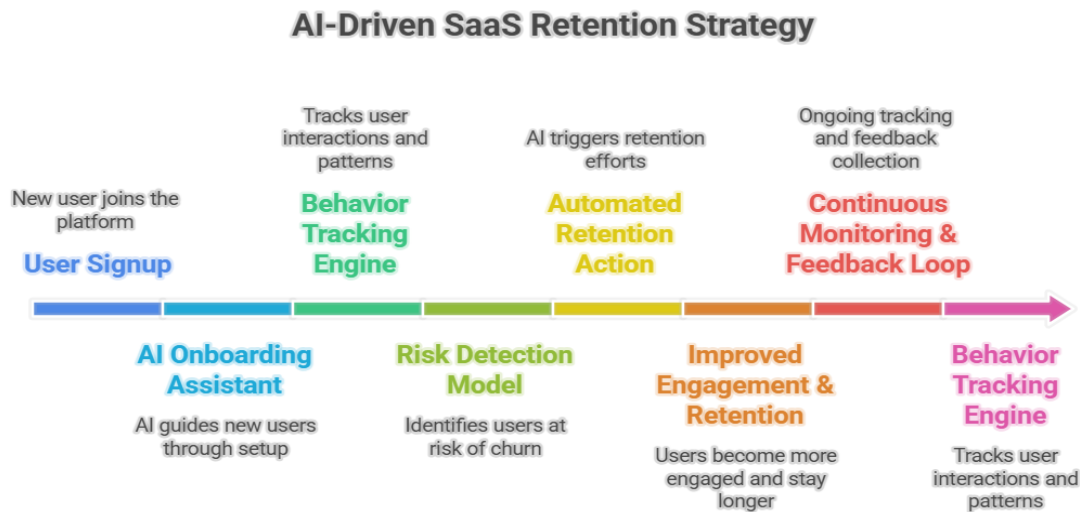
### Churn Distribution by Customer Tenure



## Churn Contribution by Contract Type



## AI-Driven SaaS Retention Workflow



## 6. Root Cause Analysis

| Root Cause                       | Evidence                               | Business Impact               |
|----------------------------------|----------------------------------------|-------------------------------|
| Weak onboarding & low engagement | 41.95% churn within first 6 months     | Early-stage customer drop-off |
| Low customer commitment          | 88.55% churn from month-to-month users | Retention instability         |
| Competitive pressure             | 45% churn linked to competitors        | Revenue & market share loss   |

## 7. Proposed AI-Driven Retention Strategy

The proposed framework introduces AI-enabled touchpoints across the customer lifecycle:

- AI onboarding assistant for guided setup and activation
- Behavioral monitoring to identify engagement drops
- Predictive churn-risk detection for proactive intervention
- Automated retention actions such as personalized nudges and support escalation

## 8. Prioritization & Validation

The onboarding assistant was prioritized as the highest-impact initiative due to the concentration of churn in the first 6 months. Proposed workflows would be validated through A/B testing focused on onboarding completion, engagement improvement, and churn reduction.

## 9. Conclusion

The analysis demonstrates that churn is highly concentrated among early-stage and month-to-month users, while competitor-driven switching remains the largest risk factor. The proposed AI-enabled retention workflow aims to improve onboarding, strengthen engagement, and proactively identify churn risks to improve long-term customer retention.